~~14/08/2026 → refer pdf~~

24/08/2026 Inference procedure under GM framework.

$\underset{n\times 1}{Y} = \underset{n\times p}{X}\underset{p\times 1}{\beta} + \underset{n\times 1}{\varepsilon}$, $\varepsilon \sim N_n(0, \sigma^2 I_n)$, $\rho(X) = p$

$H_0 : \underset{r\times p}{L^T}\beta = \underset{r\times 1}{\theta_0}$ v/s $H_1 : L^T\beta \neq \theta_0$ $(\rho(L) = r \leq p)$

* Wald's test: (Based on the BLUE for $L^T\beta = 0$)

say $\hat{\theta}$ = BLUE for $\theta = L^T\hat{\beta} \sim N_r(L^T\beta, \sigma^2 L^T(X^TX)^{-1}L)$

$\hat{\beta} = (X^TX)^{-1}X^TY$

$\hat{\sigma}^2 = MSE = \dfrac{Y^T(I-P_x)Y}{n-p}$

→ Test statistic (T^2 statistic):

$W = (\hat{\theta} - \theta_0)^T(\widehat{Var}(\hat{\theta}))^{-1}(\hat{\theta} - \theta_0)$

$= \dfrac{1}{\hat{\sigma}^2}(L^T\hat{\beta} - \theta_0)^T(L^T(X^TX)^{-1}L)^{-1}(L^T\hat{\beta} - \theta_0) \overset{d}{\underset{H_0}{=}} r\,\mathbb{F}_{r,n-p}$

→ $(L^T\hat{\beta} - \theta_0)^T(\sigma^2 L^T(X^TX)^{-1}L)^{-1}(L^T\hat{\beta} - \theta_0) \sim \chi^2_r\left(\frac{\lambda}{2}\right)$

where $\lambda = \dfrac{1}{\sigma^2}(L^T\beta - \theta_0)^T(L^T(X^TX)^{-1}L)^{-1}(L^T\beta - \theta_0)$.

$(\sigma^2 L^T(X^TX)^{-1}L)^{-1/2}(L^T\hat{\beta} - \theta_0) \sim N(V^{-1/2}(L^T\beta - \theta_0), I)$

$(V = \sigma^2 L^T(X^TX)^{-1}L)$

→ In general, $\dfrac{W}{r} \sim \mathbb{F}_{r,n-p}\left(\frac{\lambda}{2}\right)$

Power of the test $\uparrow 1$ as $\lambda \uparrow \infty$

→ Using the expression for the constrained LSE of β, under the constraint $L^T\beta = \theta_0$, we observed.

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$SSE_{reduced} - SSE_{full}$

$= (L^T\hat{\beta} - \theta_0)^T (L^T (X^T X)^{-1} L)^{-1} (L^T \hat{\beta} - \theta_0)$

Corresponding test statistic

$$= \frac{\dfrac{SSE_{reduced} - SSE_{full}}{df(\ \text{''}\ ) - df(\ \text{''}\ )}}{MSE_{full}}$$

$$= \frac{W}{(n-(p-r))-(n-p)} = \frac{1}{r} W$$

* Let Θ be the parameter space.
And $H_0: \theta \in \Theta_0$ specifies the null hypothesis where $\Theta_0 \subset \Theta$

(Likelihood ratio)

$$LR \text{ statistic} = \frac{\sup_{\theta \in \Theta_0} l(\theta \mid data)}{\sup_{\theta \in \Theta} l(\theta \mid data)}$$

→ In the present context:

$\cancel{\Theta_0 = \{\beta \in \mathbb{R}^p : L^T \beta = \theta_0\}, \ \Theta = \mathbb{R}^p}$

$\Theta = \mathbb{R}^p \times \mathbb{R}_+ \quad (\beta, \sigma^2)$

$\Theta_0 = \{\beta \in \mathbb{R}^p : L^T\beta = \theta_0\} \times \mathbb{R}_+$

$$\text{Denominator} = \sup_{\substack{\beta \in \mathbb{R}^p \\ \sigma^2 > 0}} \frac{1}{(\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \|Y - X\beta\|^2\right)$$

$$= \sup_{\sigma^2 > 0} \frac{1}{(\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \|Y - X\hat{\beta}\|^2\right) \quad (\hat{\beta} = LSE \text{ of } \beta)$$

$$\underset{\sigma}{\operatorname{argmax}} = \frac{1}{n} \|Y - X\hat{\beta}\|^2 = \tilde{\tilde{\sigma}}^2$$

$$\frac{1}{((\tilde{\tilde{\sigma}})^2)^{n/2}} \exp(-n/2)$$

Numerator $= \sup\limits_{\substack{\beta: L^T\beta=\theta_0 \\ \sigma^2>0}} \dfrac{1}{(\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\|Y-X\beta\|^2\right)$

Fix $\sigma^2 \Rightarrow \tilde{\beta} = \underset{\beta: L^T\beta=\theta_0}{\operatorname{argmin}} \|Y-X\beta\|^2$ = constrained LSE of β obtained earlier

$$\tilde{\sigma}^2 = \frac{1}{n}\|Y-X\tilde{\beta}\|^2$$

$\therefore$ Numerator $= \dfrac{1}{(\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\|Y-X\tilde{\beta}\|^2\right)$

$$= \frac{1}{(\tilde{\sigma}^2)^{n/2}} \exp\left(-\frac{n}{2}\right)$$

$\therefore LR = \dfrac{\frac{1}{(\tilde{\sigma}^2)^{n/2}} \exp\left(-\frac{n}{2}\right)}{\frac{1}{(\hat{\sigma}^2)^{n/2}} \exp\left(-\frac{n}{2}\right)} = \left(\dfrac{\|Y-X\hat{\beta}\|^2}{\|Y-X\tilde{\beta}\|^2}\right)^{n/2}$

LRT: Reject H_0 if $\dfrac{\|Y-X\tilde{\beta}\|^2}{\|Y-X\hat{\beta}\|^2} - 1$ is large.

$$= \frac{\|Y-X\tilde{\beta}\|^2 - \|Y-X\hat{\beta}\|^2}{\|Y-X\hat{\beta}\|^2}$$

$$= \frac{SSE_{red.} - SSE_{full}}{SSE_{full}}$$

Note: $-2\log LR \xrightarrow{d} \chi^2_r$

* Fundamental theorem of least squares:

$Y \sim N_r(X\beta, \sigma^2 I)$, $\beta \in \mathbb{R}^p$, $\underset{n\times p}{X}$

$\rho(X) = r \leq p$

$\hat{\beta}$ is any solution of $X^TX\beta = X^TY$

$SSE = Y^T(I-P_x)Y$

- 1st fundamental theorem : $SSE = \min_{\beta \in \mathbb{R}^p} ||Y - X\beta||^2$ &

$$SSE \sim \sigma^2 \chi^2_{n-p}$$

- 2nd fundamental theorem : Let L be a $p \times m$ matrix and $\mathcal{C}(L) \subset \mathcal{C}(X^T)$. Let $SSE_{red} = \min_{\beta : L^T\beta = \theta_0 (\in \mathbb{R}_m)} ||Y - X\beta||^2$

Then (i) SSE & $SSE_{red} - SSE$ are independently distributed
(ii) $SSE_{red} - SSE \sim \chi^2_m(\delta)$ for some non centrality parameter δ.

$$\delta = 0 \text{ if } L^T\beta = \theta_0.$$

- 3rd fundamental theorem : Let $\underset{n\times p}{X} = [X_{n\times q} : X_{n \times (p-q)}]$

Let $r_1 = \rho(X_1)$. Define $U = \left[\dfrac{\min_{\gamma \in \mathbb{R}^q} ||Y - X_1\gamma||^2}{\min_{\beta \in \mathbb{R}^p} ||Y - X\beta||^2}\right]^{-1}$

Then, $U \sim Beta\left(\frac{q-r}{2}, \frac{n-q-r+r_1}{2}\right)$

$\Rightarrow$ How to find BLUE?
$Y = X\beta + \varepsilon$, $\varepsilon \sim (0, \sigma^2 I)$
If $l^T\beta$ is estimable, then $l^T\hat{\beta}$ is the BLUE for $l^T\beta$ where $\hat{\beta}$ is any ~~least~~ LSE of β.

Ex: $Y \sim N_3\left(\begin{pmatrix} \alpha_1 \\ \alpha_1 + \alpha_2 \\ \alpha_1 + \alpha_3 \end{pmatrix}, \sigma^2 I_3\right)$. Find BLUE of $\alpha_2 - \alpha_3$ if it exists.

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$$X = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}$$

⇒ ANOVA (Analysis of variance):

a) Different levels of a factor A = a

b) " " " " " B = b

$Y \leftarrow$ response

Treatment = combination of diff levels of the factors (i, j).

$i = 1, \dots, a \quad , \quad j = 1, \dots, b$

$Y_{i,j,k}$ = approx corr. to k-th replicate of ith level of factor A & j-th level of factor B.

$k = 1, \dots, n_{ij} \rightarrow$ # of experimental units corr. to treatment (i, j).

$E(Y_{ijk}) = \mu_{ij} \quad , \quad i = 1, \dots, a \quad , \quad j = 1, \dots, b$